

AstraPay Payment Profitability

Business question

Determine why AstraPay's contribution profit per successful transaction declined between the January–February 2025 baseline period and the May–June 2025 current period, despite growth in transaction attempts and attempted payment volume.

Using transaction-level evidence that has been validated for completeness, uniqueness, accuracy, timeliness and join integrity, quantify the effects of merchant pricing mix, payment routing and processing costs, transaction latency, fraud controls, FX conversion, settlements and chargebacks.

Executive KPI:

Contribution profit per successful transaction : Total contribution profit in USD ÷ Distinct successful transactions

where : Contribution profit = Merchant fee revenue– payment processing cost– chargeback cost ± realised FX impact

Supporting KPI:

KPI	Definition	Purpose
Transaction attempts	Distinct transaction IDs created during the period	Measures payment demand
Attempted volume	Sum of valid attempted amounts converted to USD	Measures requested monetary volume
Successful transactions	Distinct transactions with status = 'SUCCESS'	Denominator for unit economics
Successful volume	Valid USD amount of successful transactions	Measures processed value
Success rate	Successful transactions ÷ attempts	Measures payment conversion
Fee revenue per success	Merchant fees in USD ÷ successful transactions	Tests the pricing hypothesis
Processing cost per success	Route costs in USD ÷ successful transactions	Tests the routing-cost hypothesis
Chargeback cost per success	Chargeback amounts in USD ÷ successful transactions	Tests the loss hypothesis
Fraud decline rate	Fraud declines ÷ fraud-evaluated transactions	Measures fraud-control strictness
Processing latency	Median and 95th-percentile processing milliseconds	Tests operational degradation

Scope

- Payment attempts created from **1 January through 30 June 2025**
- Baseline: **January–February**
- Transition: **March–April**
- Current period: **May–June**
- India, Singapore and the UAE
- Retail and SME customers
- All merchant categories, pricing plans, channels, routes and providers
- Payment attempts and their linked events, fraud decisions, settlements and chargebacks

Key assumptions: Fee amount is merchant revenue in settlement currency. Route fixed fees are in USD. MID rates convert transaction values. SETTLEMENT rates convert settlement values. Missing values are flagged and are not treated as zero.

Decisions this analysis must support

Management must decide which merchants need pricing changes, which routes should receive less traffic, whether fraud model v3.0 needs controlled testing, which data defects block reporting, and which actions should be completed during the next 60 days

Hypotheses and Supporting Evidence

The decline in profitability may be driven by commercial pricing and transaction mix, routing and processing operations, fraud, chargebacks and foreign-exchange costs, or data-quality and reporting issues. Each hypothesis should be assessed using the following evidence.

H1 : Commercial Pricing and Mix

From the CFO's perspective, profitability may be declining because fee revenue per successful transaction is falling, lower-priced plans are gaining a larger share of transactions, or the affected plans are generating low or negative contribution margins. This hypothesis would be weakened if fee revenue per successful transaction remains stable after controlling for differences in merchant and transaction mix.

H2 : Routing and Processing Operations

The Head of Payments may investigate whether expensive processing routes are handling a greater share of transactions, processing cost per successful transaction is increasing, latency is rising on affected routes, or payment success rates are falling. This hypothesis would be weakened if route mix, processing costs, and latency remain stable while profitability continues to decline.

H3 : Fraud, Chargebacks and Foreign Exchange

Risk and Treasury teams may examine whether fraud-decline rates increased following a model change, chargeback or foreign-exchange costs per successful transaction rose, or particular risk groups account for the deterioration. This hypothesis would be weakened if risk losses and foreign-exchange costs remain stable or improve, or if fraud-related changes do not coincide with the decline.

H4 : Data Quality or Reporting Artefact

The Data Head may assess whether the reported decline results from data-quality or reporting problems. Evidence supporting this hypothesis would include a material change in the reported decline after correcting deduplication, settlement aggregation, foreign-exchange-rate selection, or temporal joins. The hypothesis would be weakened if the decline remains materially unchanged after all identified data defects and unsafe joins have been corrected.